Euclid, Book I, Proposition I.12

To Draw a Perpendicular to a Given Straight Line from a Point Outside It

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1 Introduction

Euclid's Proposition I.12 asks for the construction of a perpendicular to a given straight line from a given point lying outside that line.

Classically, Euclid obtains the perpendicular by choosing a point on the opposite side of the given line, drawing a circle centered at the external point, and using the two intersections of the circle with the given line. The segment joining these intersections is bisected by Proposition I.10, and Proposition I.8 is then applied to prove that the line joining the external point to the midpoint is perpendicular to the given line.

The reconstruction over the Hilbert foundation follows a different synthetic route. No circle-line intersection theorem is used. Instead, a second point is constructed on the opposite side of the given line so that two distinct points of the line are equidistant from the original external point and the constructed point.

This leads naturally to the perpendicular-bisector configuration. The required midpoint property is isolated in the general theorem

```
hilbert_equidistant_line_bisects_segment
```

which states that if two distinct points of a line are equidistant from two points lying on opposite sides of that line, then the intersection of the joining segment with the line bisects that segment.

A second auxiliary result,

```
hilbert_collinear_angle_transport
```

encapsulates the angle-order argument needed in the proof of this perpendicular-bisector theorem.

Consequently, Proposition I.12 provides another example of the role of the Book Zero layer: a comparatively elaborate geometric configuration is separated into reusable synthetic lemmas before the final Euclidean proposition is formulated.

2 The Classical Configuration

Euclid's twelfth proposition asks for a perpendicular to a given straight line from a point outside it.

Let AB be the given straight line and let C be the given point outside it.

2.1 Final Configuration

The objective is to determine a point H on the straight line AB such that

$$CH \perp AB.$$

Euclid constructs two points E and G on AB satisfying

$$CE \cong CG,$$

and then bisects EG at H.

The final proof configuration, shown later in Figure 4, satisfies

$$E - H - G, \qquad EH \cong HG, \qquad CE \cong CG.$$

2.2 Construction Algorithm

The construction uses Proposition I.10 as an already established construction primitive.

2.2.1 Step 1. Given Line and External Point

Let AB be the given straight line and let C be a point not lying on AB; see Figure 1.

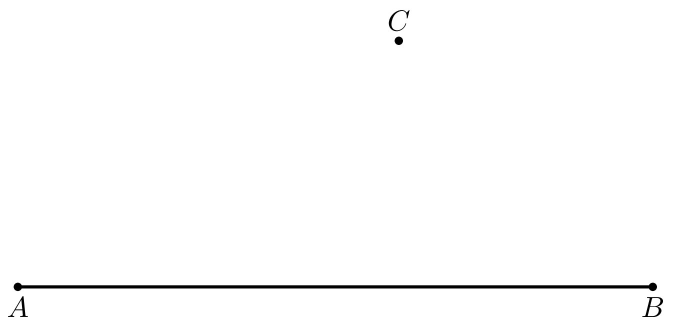

Figure 1: Input data for Proposition I.12: a straight line AB and a point C outside it.

2.2.2 Step 2. Choose a Point on the Opposite Side

Choose a point D on the opposite side of the straight line AB from C, as in Figure 2.

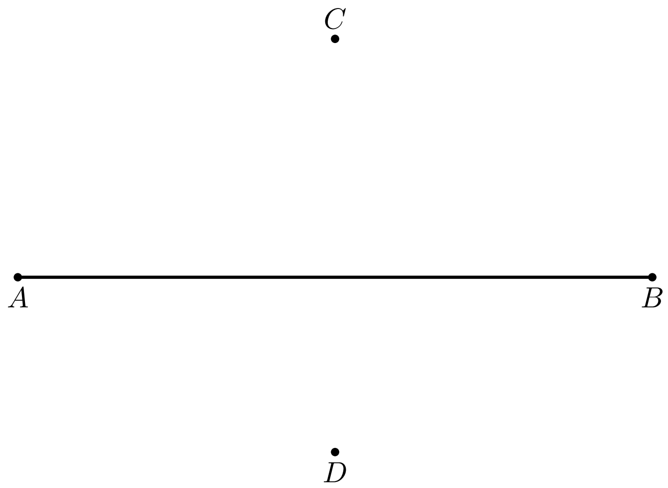

Figure 2: Choose D on the side of the line AB opposite C. This side-of-line information is essential for the later intersection argument.

2.2.3 Step 3. Draw the Circle Centered at C

Draw the circle centered at C and passing through D; the resulting intersection configuration is shown in Figure 3.

Let the circle meet the given straight line at E and G.

Since E and G lie on the same circle centered at C,

$$CE \cong CG.$$

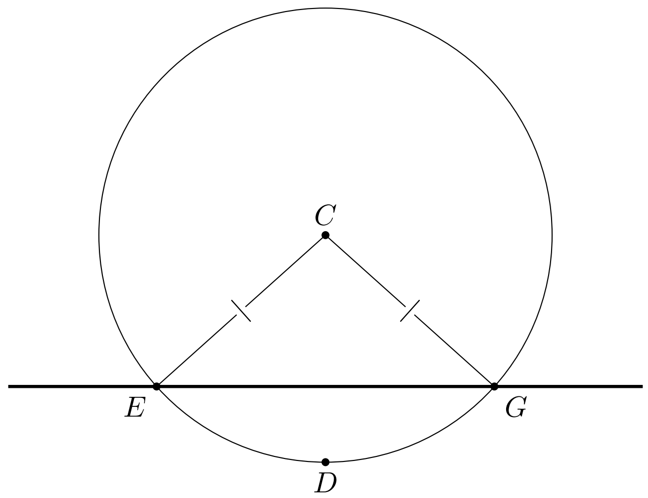

Figure 3: The circle centered at C through D meets the given line at two distinct points E and G, so $CE \cong CG$. The existence of two such intersections is a genuine geometric existence issue, not something proved by the picture itself.

2.2.4 Step 4. Bisect EG

Apply Proposition I.10 to bisect the segment EG at H.

Thus

$$E - H - G$$

and

$$EH \cong HG.$$

2.2.5 Step 5. Join CH

Join the external point C to the midpoint H; the completed proof configuration is Figure 4.

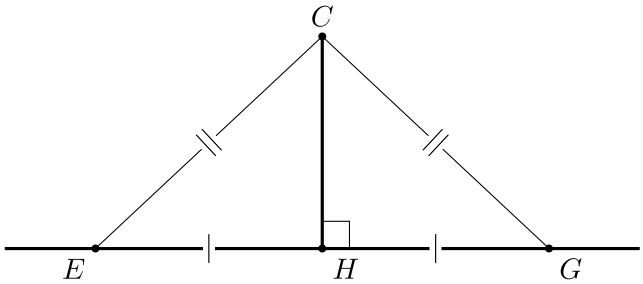

Figure 4: Completed Euclidean proof configuration. The midpoint relation $EH \cong HG$, the radii $CE \cong CG$, and the common side CH give SSS for triangles CEH and CGH.

The two triangles EHC and GHC now satisfy

$$EH \cong HG, \qquad HC \cong HC, \qquad CE \cong CG.$$

By Proposition I.8,

$$\angle EHC \cong \angle GHC.$$

Since E, H, G are collinear, these are adjacent angles. By Definition 10 they are right angles.

Therefore

$$CH \perp AB.$$

The required perpendicular has been constructed.

Diagrammatic audit. The construction sequence is retained only where the geometry genuinely changes: the input data, the opposite-side choice, the circle intersection, and the completed SSS configuration. The former midpoint-only snapshot is omitted because it adds only the label H to the preceding circle diagram. Conversely, the circle-intersection figure is essential: Wilkins explicitly notes that the existence of two distinct points E, G requires a tacit topological intersection principle. The picture records those intersections but does not prove their existence.

3 Classical Proof

Euclid's construction is obtained from the configuration described above.

Choose a point D on the opposite side of the given line AB from the external point C. Draw the circle centered at C through D, and let it meet AB at E and G. Then

$$CE \cong CG.$$

Apply Proposition I.10 to bisect EG at H. Hence

$$E - H - G \qquad \text{and} \qquad EH \cong HG.$$

Now compare the triangles

$$\triangle EHC \qquad \text{and} \qquad \triangle GHC.$$

They satisfy

$$EH \cong HG, \qquad HC \cong HC, \qquad CE \cong CG.$$

By Proposition I.8, the triangles are congruent by SSS, and therefore

$$\angle EHC \cong \angle GHC.$$

Since E, H, G are collinear, these congruent adjacent angles are right angles. Consequently

$$CH \perp AB.$$

The classical dependency pattern is therefore

$$\text{circle construction} \longrightarrow CE \cong CG \longrightarrow \text{I.10} \longrightarrow EH \cong HG \longrightarrow \text{I.8} \longrightarrow \text{equal adjacent angles} \longrightarrow CH \perp AB.$$

The formal Hilbert reconstruction below proves the same proposition by a different synthetic route and deliberately avoids the circle-line intersection used by Euclid.

4 Proof Architecture of the Reconstruction

4.1 Dependency Diagram

```
Hilbert / Book Zero dependency path
-----------------------------------

hilbert_collinear_angle_transport
              |
              v
hilbert_equidistant_line_bisects_segment
              |
              v
hilbert_perpendicular_from_point_exists
              |
              v
euclid_proposition_12
```

The formal proof separates Proposition I.12 into two reusable Book Zero lemmas and one final construction theorem.

The first auxiliary result is

```
hilbert_collinear_angle_transport
```

It transports an angle congruence when one side of the angle is replaced by another point on the same supporting line. The proof handles the two possible ray configurations: direct transport along the same ray, and transport through supplementary angles when the new ray is opposite.

The second auxiliary result is

```
hilbert_equidistant_line_bisects_segment
```

Suppose two distinct points A and B lie on a line b, while P and Q lie on opposite sides of b, and

$$AP \cong AQ, \qquad BP \cong BQ.$$

If the segment PQ meets b at H, then

$$PH \cong HQ.$$

The logical structure is therefore

```
hilbert_collinear_angle_transport
        |
hilbert_equidistant_line_bisects_segment
        |
hilbert_perpendicular_from_point_exists
        |
euclid_proposition_12
```

The main construction theorem

```
hilbert_perpendicular_from_point_exists
```

uses the perpendicular-bisector result to obtain the midpoint H of PQ. It then chooses a point R on the given line distinct from H, applies Hilbert SSS to the triangles RHP and RHQ, and uses vertical angles together with the definition of a right angle.

Finally,

```
euclid_proposition_12
```

is a thin interface theorem exposing this general Hilbert result as Euclid's Proposition I.12.

5 Hilbert Reconstruction

The Hilbert reconstruction preserves the geometric objective of Proposition I.12 but does not reproduce Euclid's circle construction.

Let A and B be distinct points of the given line b, and let P be the prescribed point outside b.

The first task is to construct a point Q on the opposite side of b from P such that

$$AP \cong AQ, \qquad BP \cong BQ.$$

To obtain Q, extend PA through A to a point S. Thus P and S lie on opposite sides of b.

Using Hilbert angle construction at A, construct a ray on the side of b containing S whose angle with AB is congruent to the angle formed by AP and AB. On this ray lay off a segment congruent to AP. Denote its endpoint by Q; see Figure 5.

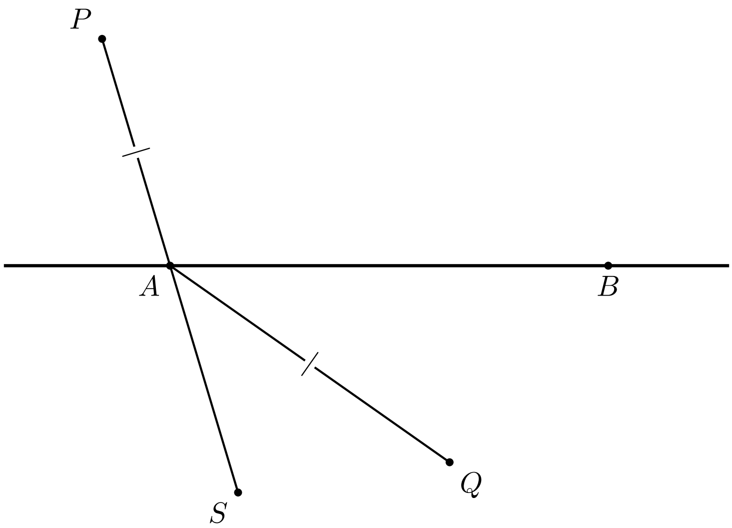

Figure 5: Hilbert construction of the reflected point Q. The line PA is extended through A to S; angle construction chooses the direction of AQ, and segment layoff gives $AQ \cong AP$.

The construction gives

$$AP \cong AQ.$$

The two triangles

$$\triangle APB \qquad \text{and} \qquad \triangle AQB$$

now have

$$AP \cong AQ, \qquad AB \cong AB,$$

and the included angles at A are congruent. By SAS,

$$BP \cong BQ.$$

Moreover, the construction places Q on the opposite side of b from P. Hence the segment PQ intersects b at a point H. The resulting configuration is shown in Figure 6.

At this point the general Book Zero theorem

```
hilbert_equidistant_line_bisects_segment
```

applies. Since

$$AP \cong AQ, \qquad BP \cong BQ,$$

and A and B are distinct points of b, the intersection point H bisects PQ:

$$PH \cong HQ.$$

Thus the difficult incidence and angle-order argument has been encapsulated before Proposition I.12 itself is completed.

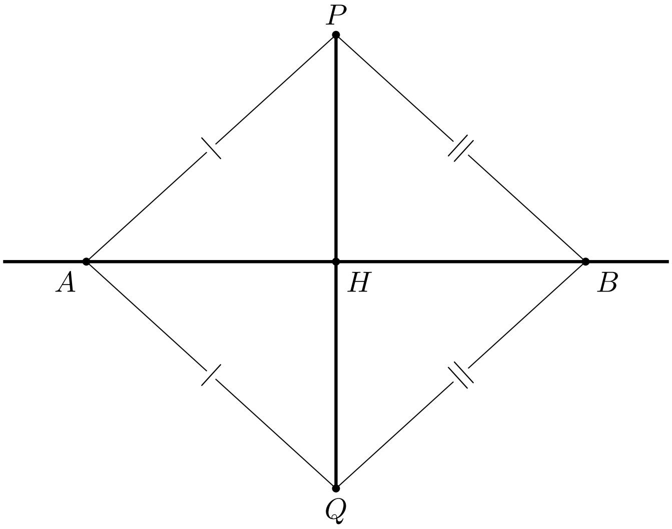

Figure 6: Hilbert perpendicular-bisector configuration. The constructed equalities $AP \cong AQ$ and $BP \cong BQ$, together with the intersection $P - H - Q$, allow the Book Zero bisector theorem to conclude $PH \cong HQ$.

To obtain the right angle, choose one of A and B distinct from H and call it R. Since R is one of the two equidistant points,

$$RP \cong RQ.$$

Together with

$$PH \cong HQ \qquad \text{and} \qquad RH \cong RH,$$

Hilbert SSS applied to

$$\triangle RHP \qquad \text{and} \qquad \triangle RHQ$$

gives congruent angles at H.

Since $P - H - Q$, these congruent angles determine a perpendicular to b at H. In the formal development this last step is expressed using vertical angles and the predicate

```
HilbertRightAngle
```

The reconstruction therefore replaces Euclid's use of a circle and Propositions I.10 and I.8 by a Hilbert construction based on angle construction, segment layoff, SAS, the perpendicular-bisector theorem, and SSS.

The resulting geometric object is nevertheless the same: a point H on the given line such that the line through P and H is perpendicular to the given line.

6 Lean Formalization

The Lean formalization separates the reusable geometric content from the final statement of Proposition I.12.

The main construction theorem is

```
theorem hilbert_perpendicular_from_point_exists
```

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
(hPbase : not HilbertIncidence.OnLine P base) :
exists H R : Geo.Point,
HilbertIncidence.OnLine H base /\
HilbertIncidence.OnLine R base /\
HilbertRightAngle Geo R H P
```

Here H is the foot of the perpendicular and R is a second point of the given line used to determine its direction at H.

The presence of R in the conclusion is important. The intersection point H may coincide with either A or B, so neither of the original points can be fixed in advance as the first point in the right-angle predicate. Since

$$A \neq B,$$

at least one of them is distinct from H and can therefore be chosen as R.

The formal proof first constructs a point Q on the opposite side of the given line from P. The construction establishes

$$AP \cong AQ$$

and, by SAS,

$$BP \cong BQ.$$

The opposite-side relation supplies an intersection point H with

$$P - H - Q.$$

The reusable Book Zero theorem

```
hilbert_equidistant_line_bisects_segment
```

then gives

$$PH \cong HQ.$$

The remaining argument selects R, applies Hilbert SSS to the triangles RHP and RHQ, and converts the resulting angle congruence into a right angle by means of vertical angles.

The Euclidean proposition itself is exposed by the wrapper

```
theorem euclid_proposition_12
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
```

```
(hPbase : not HilbertIncidence.OnLine P base) :
exists H R : Geo.Point,
HilbertIncidence.OnLine H base /\
HilbertIncidence.OnLine R base /\
HilbertRightAngle Geo R H P := by

exact
hilbert_perpendicular_from_point_exists
Geo
A B P
base
hAB
hAbase
hBbase
hPbase
```

Thus the theorem carrying Euclid's name is deliberately only a thin wrapper. The geometric work is performed by the Hilbert construction theorem and by the reusable results already established in the Book Zero layer.

7 Relationship with Earlier Propositions

7.1 Proposition I.8

Euclid uses Proposition I.8 at the final stage of the classical proof. Once the two radii from C and the two half-segments at H are known to be congruent, SSS gives congruent adjacent angles and hence the perpendicular.

The Hilbert reconstruction also uses SSS, but in a different configuration: the triangles RHP and RHQ arise from the perpendicular-bisector construction rather than from Euclid's circle.

7.2 Proposition I.10

In Euclid's proof, Proposition I.10 is the construction that bisects the chord EG.

The Hilbert reconstruction does not call I.10. Instead, the midpoint property

$$PH \cong HQ$$

is derived from the more general theorem `hilbert_equidistant_line_bisects_segment`.

Thus the Euclidean midpoint construction is replaced by a reusable perpendicular-bisector principle.

7.3 Proposition I.11

Proposition I.11 constructs a perpendicular at a point already lying on the given line. Proposition I.12 solves the complementary problem: the prescribed point lies outside the line and the foot of the perpendicular must itself be constructed.

Together I.11 and I.12 complete the two elementary perpendicular construction problems needed later in Book I.

8 Hilbert and Book Zero Components Used

8.1 `hilbert_collinear_angle_transport`

This helper transports angle congruence when one endpoint on a supporting line is replaced by another collinear point. It handles both the same-ray and opposite-ray configurations.

8.2 `hilbert_equidistant_line_bisects_segment`

This is the central reusable theorem extracted during the reconstruction of I.12. Two distinct points of a line, each equidistant from P and Q, force the intersection of PQ with that line to bisect PQ.

8.3 `hilbert_perpendicular_from_point_exists`

This theorem assembles the construction: it produces the opposite-side point, obtains the midpoint from the equidistant-line theorem, chooses a second point of the base line, and proves the resulting angle is right.

8.4 Hilbert SSS and Vertical Angles

SSS converts the three segment congruences in the triangles RHP and RHQ into angle congruence at H. Vertical-angle congruence then puts that equality into the adjacent-angle orientation required by `HilbertRightAngle`.

9 Formalization Notes

Proposition I.12 reveals a substantial difference between Euclid's constructional language and the Hilbert/Book Zero architecture.

Euclid obtains the perpendicular through a concrete circle construction. The circle centered at the external point produces two points of the given line at equal distance from that point. Proposition I.10 supplies their midpoint, and Proposition I.8 converts the resulting three side equalities into equality of the adjacent angles at the midpoint.

The Hilbert reconstruction does not attempt to reproduce this sequence literally. In particular, no circle is introduced. Instead, the essential geometric configuration behind Euclid's construction is isolated:

$$AP \cong AQ, \qquad BP \cong BQ.$$

Thus two distinct points A and B of the given line are equidistant from P and Q. The line through A and B therefore plays the role of the perpendicular-bisector axis of PQ.

This observation leads to the theorem

```
hilbert_equidistant_line_bisects_segment
```

which extracts the reusable geometric content of the configuration. Once the segment PQ intersects the line at H, the theorem gives

$$PH \cong HQ.$$

The proposition is therefore no longer organized around a particular circle construction. It is organized around an abstract property of two equidistant points and the line determined by them.

The proof of the perpendicular-bisector theorem itself exposed another reusable operation. When an angle is known relative to one point of a line, replacing that point by another collinear point requires attention to the order of the points: the relevant ray may remain unchanged or become the opposite ray. This argument is encapsulated in

```
hilbert_collinear_angle_transport
```

rather than being repeated inside Proposition I.12.

This is characteristic of the role of Book Zero. Elementary incidence, order, congruence, ray, and angle manipulations are packaged as synthetic lemmas. Later Euclidean propositions can then operate with larger geometric units instead of repeatedly descending to the Hilbert axioms.

There is also an important difference from the preceding elementary propositions. Proposition I.12 does not collapse to a single previously available Book Zero operation. Its reconstruction required the identification of genuinely reusable intermediate geometry. In this sense, the formalization of I.12 contributed new infrastructure to Book Zero rather than merely consuming existing infrastructure.

The final theorem

```
euclid_proposition_12
```

is consequently a wrapper around

```
hilbert_perpendicular_from_point_exists
```

The wrapper records the Euclidean proposition in the formal library, while the underlying theorem records the more general Hilbert construction on which it depends.

Proposition I.12 therefore illustrates the intended division of labor in the development:

$$\begin{array}{ccl}
\text{Hilbert axioms} & \longrightarrow & \text{elementary incidence, order, and congruence theory} \\
\text{Book Zero} & \longrightarrow & \text{reusable synthetic geometric mechanisms} \\
\text{Hilbert construction theorem} & \longrightarrow & \text{existence of the required perpendicular} \\
\text{Euclid wrapper} & \longrightarrow & \text{Proposition I.12}
\end{array}$$

The result is not a literal transcription of Euclid's proof. It is a reconstruction of the same geometric theorem over the Hilbert foundation, with the intermediate reasoning reorganized into reusable components.

10 Dependency Summary

The Euclidean and Hilbert dependency structures are visibly different.

For Euclid,

$$\text{circle} \longrightarrow \text{I.10} \longrightarrow \text{I.8} \longrightarrow \text{I.12}.$$

For the reconstruction,

$$\begin{gathered}
\text{Hilbert angle construction} + \text{segment layoff} + \text{SAS} \\
\downarrow \\
AP \cong AQ, \quad BP \cong BQ, \quad P, Q \text{ on opposite sides} \\
\downarrow \\
\texttt{hilbert_equidistant_line_bisects_segment} \\
\downarrow \\
PH \cong HQ \\
\downarrow \\
\text{Hilbert SSS} + \text{vertical angles} \\
\downarrow \\
\texttt{HilbertRightAngle} \\
\downarrow \\
\texttt{hilbert_perpendicular_from_point_exists} \\
\downarrow \\
\texttt{euclid_proposition_12}.
\end{gathered}$$

The reconstruction introduces no new axiom. Its main contribution is the extraction of reusable synthetic lemmas from the geometry hidden inside the classical construction.

11 Conclusion

Proposition I.12 is an especially clear example of reconstruction rather than literal transcription.

Euclid uses a circle, Proposition I.10, and Proposition I.8. The Hilbert development identifies the underlying perpendicular-bisector geometry and proves the result without reproducing the circle-line construction.

This change of route is productive. The proof yields `hilbert_collinear_angle_transport` and, more importantly, `hilbert_equidistant_line_bisects_segment` as reusable synthetic infrastructure.

The public theorem remains Euclid I.12: from a point outside a given line, a perpendicular to that line exists. Underneath that statement, however, the formal dependency graph records a more general Hilbert construction whose intermediate results can be reused independently.

Together with I.11, Proposition I.12 completes the basic perpendicular construction block and prepares the straight-angle theory beginning with Proposition I.13.